

Adjacent product-derivative inequalities: a finite-exponent dichotomy and endpoint obstructions

Run `fa_banach_001` — candidate substantial partial result

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Abstract

Open Problem 1.10 of Marbach asks when a mean-order Sobolev norm is controlled by the norm of a pointwise product of derivatives. We completely classify the subfamily whose integer derivative orders are supported on three consecutive values: the desired inequality holds for every finite Lebesgue exponent, with an explicit constant, and fails at the supremum endpoint. We also prove the $q = 1$ estimate for every symmetric two-factor gap and give two sharp endpoint results for the first fractional pair $(0, 1)$. General tuples remain open.

1 Source problem and statement of the partial result

The source is Frédéric Marbach, *A family of interpolation inequalities involving products of low-order derivatives*, arXiv:2306.06668 [1]. Open Problem 1.10 on PDF page 4 asks for which $q \in [1, \infty]$ and integers $0 \leq k_1 \leq \dots \leq k_\kappa$, with $\bar{k} = \kappa^{-1} \sum_i k_i$, one has

$$\|D^{\bar{k}}u\|_{L^{q\kappa}(\mathbb{R})} \lesssim \|D^{k_1}u \cdots D^{k_\kappa}u\|_{L^q(\mathbb{R})}^{1/\kappa} \quad (u \in C_c^\infty(\mathbb{R}; \mathbb{R})), \quad (1)$$

with the usual fractional interpretation when $\bar{k} \notin \mathbb{N}$.

As mentioned in Remark 1.3, the usual Gagliardo–Nirenberg inequalities admit generalizations in fractional Sobolev spaces. It would be natural to investigate such generalizations of Theorem 2, especially since, as noted in Remark 1.5, even for integer values of the parameters, the product $D^{k_1}u \cdots D^{k_\kappa}u$ already behaves as a fractional Sobolev norm when $\bar{k} \notin \mathbb{N}$.

Another particularly challenging problem concerns the possibility to relax the assumptions $k_i \leq j$ of Theorem 2. A natural (weaker) assumption would be $\bar{k} \leq j$. In particular, one can wonder in which settings the following result holds (corresponding to $j = \bar{k}$ and $\theta = \theta^* = 0$).

Open problem 1.10. *Let $q \in [1, \infty]$, $\kappa \in \mathbb{N}^*$ and $0 \leq k_1 \leq \cdots \leq k_\kappa \in \mathbb{N}$. Let $\bar{k} := (k_1 + \cdots + k_\kappa)/\kappa$. When is it true that, for $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$,*

$$\|D^{\bar{k}}u\|_{L^{q\kappa}(\mathbb{R})} \lesssim \|D^{k_1}u \cdots D^{k_\kappa}u\|_{L^q(\mathbb{R})}^{\frac{1}{\kappa}}, \quad (1.13)$$

where the left-hand side should be interpreted as the fractional $\dot{W}^{\bar{k}, q\kappa}(\mathbb{R})$ semi-norm of u when $\bar{k}$ is not an integer.

As noted in Remark 1.5, positive answers to Open problem 1.10 imply Theorem 2 (up to exceptional cases) thanks to the (fractional) Gagliardo–Nirenberg inequality $\|D^j u\|_{L^p(\mathbb{R})} \lesssim \|D^m u\|_{L^r(\mathbb{R})}^\theta \|D^{\bar{k}} u\|_{L^{q\kappa}(\mathbb{R})}^{1-\theta}$ (see e.g. [3, 4] when $\bar{k} \notin \mathbb{N}$).

Unfortunately, the proofs of Section 2 do rely on the assumptions $k_i \leq j$. In particular, estimate (2.10) below is false if there exists i such that $k_i > j$. Nevertheless, *ad hoc* arguments entail that (1.13) holds for some examples, hinting that Open problem 1.10 might have positive answers.

Lemma 1.11. *For $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$,*

$$\|u\|_{\dot{W}^{\frac{1}{2}, 4}(\mathbb{R})} \lesssim \|uu'\|_{L^2(\mathbb{R})}^{\frac{1}{2}}, \quad (1.14)$$

$$\|u'\|_{L^4(\mathbb{R})} \lesssim \|uu''\|_{L^2(\mathbb{R})}^{\frac{1}{2}}, \quad (1.15)$$

$$\|u'\|_{L^6(\mathbb{R})} \lesssim \|uu'u''\|_{L^2(\mathbb{R})}^{\frac{1}{3}}. \quad (1.16)$$

Proof. Estimate (1.14) can be derived from the remark that $(u^2)' = 2uu'$. Hence

$$\|u\|_{\dot{W}^{\frac{1}{2}, 4}(\mathbb{R})} \lesssim \|u\|_{\dot{W}^{\frac{1}{2}, 4}(\mathbb{R})} \lesssim \|u^2\|_{\dot{H}^1(\mathbb{R})}^{\frac{1}{2}} = \|2uu'\|_{L^2(\mathbb{R})}^{\frac{1}{2}}, \quad (1.17)$$

Figure 1: Open Problem 1.10 and the source paper’s three examples (arXiv:2306.06668, PDF page 4).

Our main partial answer is the following exact dichotomy.

Theorem 1 (Adjacent integer orders). *Let $j \geq 1$, $a \geq 1$, and $b \geq 0$ be integers, and set $\kappa = 2a + b$. For every $1 \leq q < \infty$ and $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$,*

$$\|D^j u\|_{L^{q\kappa}} \leq (q\kappa - 1)^{a/\kappa} \|(D^{j-1}u)^a (D^j u)^b (D^{j+1}u)^a\|_{L^q}^{1/\kappa}. \quad (2)$$

For $q = \infty$, there is no finite constant C such that

$$\|D^j u\|_{L^\infty} \leq C \|(D^{j-1} u)^a (D^j u)^b (D^{j+1} u)^a\|_{L^\infty}^{1/\kappa} \quad (3)$$

for every $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$.

The derivative multiset in this theorem has a copies of $j-1$, b copies of j , and a copies of $j+1$. Its mean is j . Conversely, every mean- j multiset supported on $\{j-1, j, j+1\}$ has exactly this form. Thus Theorem 1 classifies the entire adjacent-order subfamily of the source problem.

Theorem 2 (All symmetric gaps at $q = 1$). *Let $j \geq d \geq 1$ be integers. Then every $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$ satisfies*

$$\|D^j u\|_{L^2} \leq \|D^{j-d} u D^{j+d} u\|_{L^1}^{1/2}. \quad (4)$$

Finally, the first fractional pair already shows different endpoint behavior. We use the standard Gagliardo seminorm at finite exponent and the homogeneous Hölder interpretation $[u]_{\dot{W}^{1/2, \infty}} = \sup_{x \neq y} |u(x) - u(y)|/|x - y|^{1/2}$ at infinity.

Theorem 3 (The fractional pair $(0, 1)$). *There is no constant C such that*

$$[u]_{\dot{W}^{1/2, 2}(\mathbb{R})} \leq C \|uu'\|_{L^1}^{1/2} \quad (u \in C_c^\infty(\mathbb{R}; \mathbb{R})). \quad (5)$$

At the supremum endpoint, however, every $u \in C_c^\infty(\mathbb{R}; \mathbb{R})$ satisfies

$$[u]_{\dot{W}^{1/2, \infty}} \leq 2 \|uu'\|_{L^\infty}^{1/2}. \quad (6)$$

The source proves the same pair at $q = 2$ (its Lemma 1.11). We do not settle the other finite exponents here.

2 Idea of the proof

For the adjacent family, put $f = D^{j-1} u$. The prescribed product becomes $f^a (f')^b (f'')^a$. One integration by parts writes the $L^{q\kappa}$ energy of f' as an integral containing one copy of $f f''$. Taking the a -th root of the full prescribed product and applying Hölder closes exactly: the identity $\kappa = 2a + b$ is precisely the exponent balance.

The divergence of the resulting constant as $q \rightarrow \infty$ is not an artifact. The autonomous equation $y'' = A(y)/y$ keeps $|yy''|$ bounded while its energy lets $|y'|^2$ accumulate a logarithm over a large range of values. A cutoff made in the *value variable* turns one periodic orbit into a smooth compactly supported bump without losing either feature. Separated finite differences impose the vanishing moments needed to make that bump a high derivative of another compactly supported function.

For the fractional $q = 1$ obstruction, smooth approximations to an interval indicator have bounded variation of u^2 , hence bounded $\|uu'\|_1$, but critical $H^{1/2}$ energy growing like the logarithm of the inverse transition width. At $q = \infty$, the elementary identity $(u^2)' = 2uu'$ gives the opposite conclusion after separating same-sign and opposite-sign pairs.

3 Finite exponents and symmetric gaps

Proof of the finite- q part of Theorem 1. Set

$$f = D^{j-1} u, \quad P = q\kappa, \quad G = |f|^a |f'|^b |f''|^a, \quad I = \int_{\mathbb{R}} |f'|^P.$$

Since $P \geq 2$, integration by parts gives

$$I = -(P-1) \int_{\mathbb{R}} f f'' |f'|^{P-2} \leq (P-1) \int_{\mathbb{R}} G^{1/a} |f'|^{P-2-b/a}. \quad (7)$$

Suppose first that $aq > 1$. Hölder's inequality with conjugate exponents aq and $aq/(aq-1)$ applies because

$$\left(P-2-\frac{b}{a}\right) \frac{aq}{aq-1} = P; \quad (8)$$

indeed $P = q(2a+b)$. Thus

$$I \leq (P-1) \left(\int_{\mathbb{R}} G^q \right)^{1/(aq)} I^{(aq-1)/(aq)}.$$

Consequently

$$I^{1/q} \leq (P-1)^a \|G\|_{L^q}.$$

But $I^{1/q} = \|f'\|_{L^P}^{P/q} = \|f'\|_{L^P}^\kappa$, so taking κ -th roots proves (2).

If $aq = 1$, then $a = q = 1$ and $P = 2 + b$. The residual exponent in (7) is $P - 2 - b = 0$, so the same conclusion follows directly. This also covers the only limiting case omitted from the Hölder calculation. $\square$

For $a = 1$ and $b = 0$, the argument is the familiar nonlinear Gagliardo–Nirenberg/Opial inequality. Kałamajska–Peszke give this finite-exponent inequality, in greater weighted generality, in arXiv:1104.1967 [2]. The point here is the exact repeated factor extension and its placement in the source problem.

Proof of Theorem 2. Integrating by parts d times gives the exact identity

$$\int_{\mathbb{R}} D^{j-d} u D^{j+d} u = (-1)^d \int_{\mathbb{R}} (D^j u)^2.$$

Taking absolute values and then applying the triangle inequality proves (4). $\square$

4 The supremum counterexample

We first construct the basic bad bump.

Lemma 4 (Phase-plane bump). *There is a universal $K < \infty$ and a family of nonnegative functions $f_\varepsilon \in C_c^\infty(\mathbb{R})$, $0 < \varepsilon < 1/2$, such that*

$$\|f_\varepsilon f_\varepsilon''\|_{L^\infty} \leq K, \quad \|f_\varepsilon'\|_{L^\infty}^2 \geq 2 \log \frac{1}{\varepsilon}. \quad (9)$$

Proof. Choose $A \in C^\infty((0, \infty); [-1, 1])$ such that $A(y) = 1$ for $y \leq 1$ and $A(y) = -1$ for $y \geq e$. Define

$$H_\varepsilon(y) = 2 \int_\varepsilon^y \frac{A(t)}{t} dt.$$

For small ε , H_ε is positive immediately to the right of ε , satisfies $H_\varepsilon(1) = 2 \log(1/\varepsilon)$, and has a first zero $Y_\varepsilon > e$. The solution of

$$y'' = \frac{A(y)}{y}, \quad y(0) = \varepsilon, \quad y'(0) = 0 \quad (10)$$

obeys the conserved energy

$$(y')^2 = H_\varepsilon(y). \quad (11)$$

It travels from the minimum ε to the turning point Y_ε in finite time and, by time reversal, returns smoothly to ε . Let $[0, T_\varepsilon]$ be this min-to-min period.

Choose a smooth step ρ which is zero on $(-\infty, 0]$, one on $[1, \infty)$, and flat at both endpoints, and set

$$\chi_\varepsilon(y) = y\rho\left(\frac{y - \varepsilon}{\varepsilon}\right).$$

Define $f_\varepsilon(x) = \chi_\varepsilon(y(x))$ on $[0, T_\varepsilon]$ and zero outside. Flatness of ρ at zero makes this extension C^∞ .

When $y \geq 2\varepsilon$, one has $\chi_\varepsilon(y) = y$, hence $|f_\varepsilon f_\varepsilon''| = |yy''| = |A(y)| \leq 1$. In the transition strip $\varepsilon \leq y \leq 2\varepsilon$,

$$|\chi_\varepsilon| \leq 2\varepsilon, \quad |\chi'_\varepsilon| \leq C, \quad |\chi''_\varepsilon| \leq C/\varepsilon, \quad (y')^2 = 2\log(y/\varepsilon) \leq 2\log 2, \quad |y''| \leq 1/\varepsilon.$$

The chain rule $f''_\varepsilon = \chi''_\varepsilon(y)(y')^2 + \chi'_\varepsilon(y)y''$ therefore gives a universal bound K in (9). At $y = 1$, the cutoff is the identity and (11) gives $|f'_\varepsilon|^2 = 2\log(1/\varepsilon)$. $\square$

Proof of the $q = \infty$ part of Theorem 1. For $j = 1$, take $u = f_\varepsilon$. Writing $M_\varepsilon = \|f'_\varepsilon\|_\infty$ and using Lemma 4,

$$\|f_\varepsilon^a (f'_\varepsilon)^b (f''_\varepsilon)^a\|_\infty \leq K^a M_\varepsilon^b.$$

The ratio of the left side of (3) to its product term is therefore at least

$$K^{-a/\kappa} M_\varepsilon^{1-b/\kappa} = \left(\frac{M_\varepsilon^2}{K}\right)^{a/\kappa} \longrightarrow \infty.$$

Now let $j > 1$ and put $n = j - 1$. Choose h larger than the diameter of the support of f_ε , and form the separated finite difference

$$g_\varepsilon(x) = \sum_{r=0}^n (-1)^r \binom{n}{r} f_\varepsilon(x - rh).$$

Its translated supports are disjoint. Moreover, for $0 \leq m < n$,

$$\int_{\mathbb{R}} x^m g_\varepsilon(x) \, dx = \int_{\mathbb{R}} f_\varepsilon(t) \sum_{r=0}^n (-1)^r \binom{n}{r} (t + rh)^m \, dt = 0,$$

because an n -th finite difference annihilates polynomials of degree below n . Hence there exists $u_\varepsilon \in C_c^\infty(\mathbb{R})$ with $D^n u_\varepsilon = g_\varepsilon$; explicitly, for $n \geq 1$, take

$$u_\varepsilon(x) = \frac{1}{(n-1)!} \int_{-\infty}^x (x-t)^{n-1} g_\varepsilon(t) \, dt.$$

The vanishing moments make this primitive zero beyond the support.

On each translated component, g_ε and its first two derivatives are the same binomial coefficient times the corresponding derivatives of f_ε . The coefficient cancels from the homogeneous ratio above. Thus the same divergent ratio holds with $(D^{j-1}u_\varepsilon, D^j u_\varepsilon, D^{j+1}u_\varepsilon) = (g_\varepsilon, g'_\varepsilon, g''_\varepsilon)$, completing the proof. $\square$

5 The first fractional pair

Proof of the failure in Theorem 3. Fix a smooth increasing step η from zero to one on $[0, 1]$, flat at both endpoints. For $0 < \delta < 1/8$, let u_δ be zero outside $[0, 1]$, equal to $\eta(x/\delta)$ on $[0, \delta]$, equal to one on $[\delta, 1 - \delta]$, and equal to $\eta((1 - x)/\delta)$ on $[1 - \delta, 1]$. Then $u_\delta \in C_c^\infty$ and

$$\|u_\delta u'_\delta\|_{L^1} = \frac{1}{2} \text{Var}(u_\delta^2) = 1.$$

On the other hand, $u_\delta(x) = 1$ for $x \in [2\delta, 1/2]$ and $u_\delta(y) = 0$ for $y \leq 0$. Consequently

$$\begin{aligned} [u_\delta]_{\dot{W}^{1/2,2}}^2 &\geq \int_{2\delta}^{1/2} \int_{-x}^0 \frac{1}{(x-y)^2} dy dx \\ &= \frac{1}{2} \int_{2\delta}^{1/2} \frac{dx}{x} = \frac{1}{2} \log \frac{1}{4\delta} \rightarrow \infty. \end{aligned}$$

This contradicts any uniform form of (5). □

Proof of the positive endpoint in Theorem 3. Let $M = \|uu'\|_\infty$. Since $(u^2)' = 2uu'$, the function u^2 is $2M$ -Lipschitz. If $u(x)$ and $u(y)$ have the same sign, then

$$|u(x) - u(y)|^2 \leq |u(x)^2 - u(y)^2| \leq 2M|x - y|.$$

If their signs differ, continuity gives z between x and y with $u(z) = 0$. Therefore

$$|u(x) - u(y)|^2 = (|u(x)| + |u(y)|)^2 \leq 2(|u(x)|^2 + |u(y)|^2) \leq 4M|x - y|.$$

Taking the supremum over $x \neq y$ proves (6). □

6 Verification, limitations, and novelty

The deterministic checker evaluates Theorem 1 on five Schwartz-function parameter sets, checks the signed integration-by-parts identity in Theorem 2 for five gaps through $d = 5$, and samples the phase-plane construction for four values of $\log(1/\varepsilon)$. All checks pass. They are sanity tests, not parts of the proof.

The full source problem remains open. In particular, the present argument does not control wider gaps at $q > 1$: further integrations by parts generate intermediate derivative monomials absent from the prescribed product. Apart from the two endpoints in Theorem 3, general noninteger mean orders are also untouched.

The bounded novelty check covered the run's registry, solution, attempt, and proof-gap indexes; the exact source id/title and Open Problem 1.10; searches for the displayed adjacent and supremum inequalities; the nonlinear Gagliardo–Nirenberg paper of Kałamańska–Peszke; and the later higher-order Opial paper of Klaassen [3]. The known $a = 1, b = 0$ finite-exponent case is explicitly credited above. No located source states the repeated-factor adjacent dichotomy, the compactly supported supremum obstruction, or their identification with Marbach's problem. Novelty is therefore plausible but provisional.

Human-review recommendation. Review as a substantial partial result. The most sensitive points are the uniform transition-strip estimate in Lemma 4 and the moment argument ensuring that the finite-difference primitive is compactly supported. The finite- q proof and both fractional endpoint arguments are elementary.

References

- [1] F. Marbach, *A family of interpolation inequalities involving products of low-order derivatives*, arXiv:2306.06668 (2023).
- [2] A. Kałamajska and J. Peszek, *On some nonlinear extensions of the Gagliardo–Nirenberg inequality with applications to nonlinear eigenvalue problems*, arXiv:1104.1967; Asymptotic Analysis 77 (2012), 169–196.
- [3] C. A. J. Klaassen, *Sharp unifying generalizations of Opial’s inequality*, arXiv:2312.05150; J. Inequal. Appl. 2023, 153.